ScamShield US: An Agentic Retrieval-Augmented Generation System for Consumer Fraud Education

Final Project Group

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Problem and Motivation

Consumers frequently receive phishing, government and banking impersonation, investment, and identity-theft themed SMS messages. This proposal describes *ScamShield US*, an educational agentic retrieval-augmented generation (RAG) assistant that (a) classifies suspicious messages and (b) answers user questions using official public guidance. We selected this problem because it combines classical natural language processing (NLP) classification with modern transformer retrieval and generation, maps to real consumer harm, and fits the course stack constraint of Amazon Web Services (AWS) Elastic Compute Cloud (EC2) plus Amazon Simple Storage Service (S3) with open-source models only (no Amazon Bedrock or paid large language model APIs).

Dataset

We use two public SMS corpora: the UCI SMS Spam Collection (Almeida et al., n.d.) and the Super SMS Dataset (SMS Spam Research, n.d.). Raw files are stored under a course S3 prefix. We merge, deduplicate, and create train, validation, and test splits. Separately, we seed at least 20 URL-backed official knowledge documents (for example, Federal Trade Commission, Cybersecurity and Infrastructure Security Agency, Social Security Administration, and Internal Revenue Service guidance) for RAG.

NLP Methods From the Course Concept List

We use both classical and pretrained transformer methods from the course concept list:

- Rule and classical models: term frequency–inverse document frequency (TF-IDF) features with logistic regression and linear support vector machines (Traditional NLP).

- Pretrained transformers: DistilBERT for advanced classification (Sanh et al., 2019); MiniLM embeddings for dense retrieval (Reimers & Gurevych, 2019); Qwen2.5-1.5B-Instruct as a local decoder model for grounded answers (Qwen Team, 2024).
- Hybrid cascade: keep Traditional predictions when confident; escalate weak cases to DistilBERT.
- Retrieval-augmented generation: hybrid BM25 and FAISS (Johnson et al., 2019) with top_k = 5 over official guidance, then Qwen answers with citations.

We emphasize classical and transformer pretrained models over recurrent networks and autoencoders for this product-facing pipeline. Interpretability uses LIME after binary spam/ham prediction: corpus word clouds are shown first, then LIME word clouds aggregated from post-prediction local explanations (SHAP remains optional future work).

Packages

Planned packages include scikit-learn and joblib for classical classifiers; Hugging Face Transformers, PyTorch, Sentence-Transformers, FAISS, and Rank-BM25 for DistilBERT, MiniLM, Qwen, and hybrid retrieval; FastAPI, Uvicorn, and Angular for the application programming interface and user interface; boto3 as the only S3 access path; and uv for reproducible environments via *pyproject.toml* and *uv.lock*.

NLP Tasks

The project addresses (a) text classification (binary ham/spam and multi-class scam subtypes), (b) information retrieval (hybrid sparse and dense ranking), and (c) grounded answer generation for educational question answering with a decoder language model.

Evaluation Metrics

The primary metric is macro-averaged F1 because rare scam subtypes matter. We also report accuracy, weighted F1, precision, recall, and confusion matrices on a held-out test split. The hybrid cascade margin is selected on validation. RAG quality is assessed qualitatively via

cited answers in the interface and retrieval provenance.

Schedule

Table 1

Proposed Project Schedule

Week	Milestone
1	Proposal, dataset acquisition, repository skeleton, S3 layout
2	Baseline Traditional classifiers and DistilBERT training
3	Knowledge seeding, hybrid index, and Qwen RAG answers
4	API, Angular UI, evaluation tables, and dashboard visualizations
5	Report, presentation, GitHub deliverable folders, and polish

Note. Schedule is approximate and aligns with course deliverable deadlines.

References

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